DSA REFERENCE

Fast and Slow Pointers

Recognition cues, invariants, code templates, and 5-8 referenced problems.

Move two traversals at different speeds so their relative motion reveals cycles, midpoints, or phase boundaries.

VERSION 1.0 REVIEW
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STATUS HUMAN REVIEW REQUIRED

DSA / FAST-SLOW-POINTERS

DSA

Fast and Slow Pointers

CORE IDEA

Move two traversals at different speeds so their relative motion reveals cycles, midpoints, or phase boundaries.

Recognition cues

+ linked-list cycle
+ middle node
+ repeated state transition
+ constant-space sequence detection

Implementation sequence

+ guard every multi-step dereference
+ explain relative speed
+ separate detection from entry location
+ test odd and even lengths

Invariant

Both pointers follow the same deterministic next relation; only their speed or restart position differs.

Worked reasoning

After slow and fast meet in a cycle, reset one pointer to head and move both one step to find the entry.

Complexity

O(n) time and O(1) space for linked structures or deterministic state machines.

Failure modes

- [ ] unsafe fast.next.next access
- [ ] returning the meeting point as the entry
- [ ] misstating even-length midpoint
- [ ] using it when transitions are nondeterministic

Practice ladder

Fast and Slow Pointers: referenced problems

Open the official problem page, classify the pattern, write the invariant, and predict the complexity before coding.

#	Problem	Deliberate-practice prompt
1	**Linked List Cycle**	State the cue, invariant, edge case, and proof before implementation.
2	**Linked List Cycle II**	State the cue, invariant, edge case, and proof before implementation.
3	**Middle of the Linked List**	State the cue, invariant, edge case, and proof before implementation.
4	**Happy Number**	State the cue, invariant, edge case, and proof before implementation.
5	**Find the Duplicate Number**	State the cue, invariant, edge case, and proof before implementation.

Before coding

+ Pattern cue
+ State
+ Invariant
+ Boundary convention

Before finishing

+ Dry run
+ Edge case
+ Time proof
+ Space proof

PUBLISHING STATUS

REVIEW GATE: STABLE TECHNICAL. AI-produced draft. A named human must verify technical accuracy, sourcing, freshness, and editorial quality before publication. Reviewed-through date: 2026-07-18.